Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Progress update — the modelling half, and an honest *negative* result

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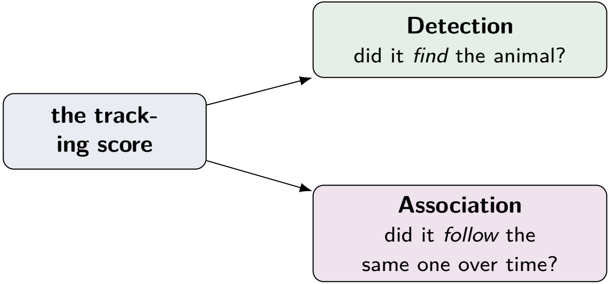
The result, in one sentence

We can **measure** SAM 3's reliability — but we **cannot predict it in advance** from label-free "distance" signals, beyond a small and obvious **animal-size** effect.

- This is a **clean, rigorous negative result** — and it is the contribution.
- It is *not* a weak-test artefact: the same test *does* pick up the size effect (shown later).
- Everything below is built, tested, and validated out-of-sample with honest statistics.

The question

Point SAM 3 at a video, type "`impala`", and it finds and follows the animal — **with no training on that species**. Can we tell, *before running it*, whether it will work for a **new animal** in a **new place** — **without labelling anything first**?



We split the question in two — that split is the intellectual core of the project.

What we've done

1. ✓ **Measured the model.** Ran frozen SAM 3 over the test videos and scored it with the *official* evaluator — a reliability number for each (species, place, time) group.
2. ✓ **Built four "distance" signals** — how *different* a new species/place is from the familiar training data: the animal (tree-of-life + appearance), the scene, and the time gap.
3. ✓ **Fitted a proper model** (score = function of the distances) and tested it **honestly out-of-sample**: hide whole species and whole places, predict them, compare to the truth — with conservative, group-aware error bars.
4. ✓ **Tried a second idea** when the first failed — maybe it is how *hard* the footage is (dark/night, cluttered, small animal), not how *novel* it is.
5. ✓ **Checked a side question** — when the animal is *absent*, does the model hallucinate one?

What the pipeline sees — a real frame

frame + ground-truth mask

visual: the animal alone

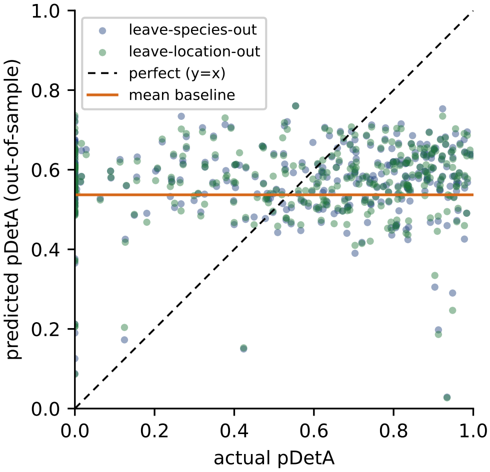
environment: animal erased

The distances are built from label-free crops of a real camera-trap frame (a collared peccary): the animal's appearance (*visual*), the scene with the animal erased (*environment*), plus its place on the tree of life and its timestamp. **No SAM 3 is used to build them.**

The main result: the distances predict *nothing*

Predict from	error	beats "guess"?
guess the average	0.298	—
+ four distances	0.298	– **no**
+ animal size	0.281	✓ a little

out-of-sample error, held-out species / places
(lower is better)



predictions collapse onto the *mean*, not the diagonal

The one “signal” was a trick of animal size

- One distance (**visual**) looked significant — but with the *wrong* sign, and it was a **confound**: “visually distinctive species do better” really means “**bigger animals are easier to spot**.”
- Add an animal-size control and the effect **vanishes** (coefficient $+0.37 \to +0.22$, now overlapping zero); size itself is the real driver.

The “difficulty” idea collapsed the same way

Dark / cluttered footage *looked* predictive — but only because it tracks animal size. On its own it predicts nothing (a strong-looking $r = -0.38$ across places shrinks to -0.13 per clip: a classic grouping illusion). **The difficulty pivot is also a null.**

Peeling it apart: only *size* survives

Predict detection from…	Beats guessing?	Gain
clip length only	– no	≈ 0
animal size only	✓ **yes**	$+0.017$
low-light only (no size)	– no	$+0.004$ (n.s.)
low-light + size	✓ yes (= size)	$+0.020$

- Size **alone** carries the whole gain; low-light **alone** does not validate.
- So the *only* label-free property that predicts detection is **object size** — small, obvious, and the very thing we introduced as a **nuisance control**.

“Following” barely differs from “finding”

- Almost every clip has a **single animal**, so the two scores are nearly identical — they agree about **94%** of the time.
- There is very little separate “following” signal for *any* feature to predict.
- So the association half is honestly a **detection story**; we report its null with those numbers as the explanation, not as a separate failed search.

The detection-vs-association split was the right question to ask — the honest finding is that, on this data, association carries almost no distinct signal.

Is “we found nothing” trustworthy?

The obvious worry

“Maybe the test just isn’t sensitive enough.”

- The **same test**, on the **same data**, *does* detect the animal-size effect out-of-sample.
- A test that **fires for size** but **stays flat for the distances** is genuinely finding *no signal* — not failing to look.

This is the key defence of the negative result.

The null holds under scrutiny

Drop any one distance — the fit and the (null) out-of-sample gain barely move. Only **adding animal size** changes anything.

Model	pseudo-R^2	beats guess?
all four distances	0.055	– no
drop any one	0.04–0.06	– no
+ animal size	0.119	✓ yes

- The four distances together explain only $\approx$ **3%** of the variance.
- No factor dominates, and every collinearity score is ≈ 1 — so it is **not** collinearity hiding a real effect.
- The null is not an artefact of one factor or one modelling choice.

Two other things we checked

Hallucinations (animal absent)

SAM 3 wrongly returns an animal about **10%** of the time. More distinctive species hallucinate *less* (a real, size-independent correlation) — but out-of-sample it **just misses** significance ($p = 0.053$). A genuine correlation, **not** a validated predictor.

The model's own confidence (explored, then set aside)

If we let SAM 3 *run first* and read its own confidence, that *does* predict its accuracy. We deliberately keep it out of the contribution: (1) it needs running the model, defeating the "predict *before* running" goal; and (2) it is near-circular (a known result). We use it only as an internal check that the pipeline can detect a strong signal.

Why a negative result is a real contribution

- **First of its kind:** no one has asked this for a promptable video *tracker*, or split it into detection vs association — novel regardless of the answer.
- **Useful for practitioners:** "far-from-training" does *not* mean "unreliable," so conservation teams should **not** trust distance-from-training as a safety check — spot-audit with a few labels instead.
- **Stronger than a weak positive:** a robust null from a demonstrably powerful test is more defensible than a borderline effect scraped past a threshold.

Where we are, and what's next

Done & validated

Measurement ✓ four distances ✓ modelling + out-of-sample validation ✓ difficulty pivot ✓ false positives ✓ ablations & variance ✓ write-up ✓

- **Write-up:** results, discussion, and figures now reflect the negative result and its explanation; the dissertation builds end-to-end.
- **In progress:** three robustness re-runs (a different visual encoder, whole-frame crops, a generic prompt) — awaiting a GPU; the null is expected to hold.
- **Honest limitation:** distances are to the *dataset's* training split, not SAM 3's true (undisclosed) training data. **Open direction:** probe whether SAM 3's own features "know" a species.

Thank you — questions welcome.